

Juan Esteban Villegas Delgado

Ph.D. Electrical Engineering
Mechatronics Engineer, M.Sc. Microsystems Engineering
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EDUCATION

Escuela de Ingeniería de Antioquia, Medellín, ANT, Colombia
Bachelor of Sciences, Mechatronics Engineering, December 2011
Thesis: “Model of solar powered water pumping systems for small distribution networks on rural arid regions in Colombia”.

Khalifa University (Masdar Institute), Abu Dhabi, AD, United Arab Emirates
Master of Science, Microsystems Engineering, May 2018
Thesis: “Studies of materials for planar integrated photonics: PDMS and amorphous Silicon”, work carried at the Masdar Institute, UAE, in collaboration with MIT.

New York University, New York City, USA & Abu Dhabi, UAE
Philosophy Doctor, Electrical Engineering, October 2023
Thesis: “Integrated Silicon Photonics for Physical Layer Security Applications”, work carried at NYU Abu Dhabi with the Center for Cyber Security at NYUAD.

PROFILE

I am a dedicated researcher and engineer with a robust background in mechatronics and electrical engineering, coupled with extensive experience in integrated electromagnetics, optics, and microwave systems. My research spans high-speed communications, sensing, and computing, focusing on designing, manufacturing, and testing innovative multimode and polarization-switching architectures in the infrared spectrum and superconductive microwave systems for quantum processing units.

Leveraging my expertise in system integration, signal processing, and photonic and microwave circuit design, I have successfully contributed to groundbreaking projects that push the boundaries of instrument performance and technological innovation. My hands-on experience with advanced micro-fabrication techniques, coupled with my proficiency in various programming languages and simulation tools, enables me to develop and optimize complex control and automated testing setups. I am experienced in managing tight schedules and resolving complex technical problems, quickly adapting to diverse academic and professional environments, and collaborating effectively with multidisciplinary teams. My excellent communication skills and experience interacting with clients and suppliers throughout the industry have honed my ability to resolve technical issues efficiently.

Proficient in Spanish, French, and English, I bring a global perspective and a high level of commitment to my work, consistently striving for excellence in all my endeavors.

TECHNICAL SKILLS

System integration and automation: Integration of automated testing setups. Development and implementation of automated data acquisition systems. Programming and deploying automation scripts for process control. Sensor integration and calibration for industrial applications.

Opto-electronic characterization: Insertion Loss and Vector analysis of optical systems, Optical Network Analysis, High speed probing and testing, DC I-V curve tracing, Design and Assembly of optical systems. Experience testing RF system in

cryogenics to temperatures below 20 mK. Experience operating and servicing dilution refrigerators, ultra-low vacuum systems and standard cryogenic equipment.

Material Characterization: Scanning Electron Microscopy (SEM), X-Ray diffraction (XRD), energy dispersive spectroscopy (EDS), atomic force microscopy (AFM), optical interferometry, Raman spectroscopy, ellipsometry, mass spectroscopy, ion chromatography and custom optical spectrometry.

Microfabrication: Physical and chemical deposition of materials (Atomic Layer Deposition (ALD), Chemical Vapor Deposition (CVD), plasma enhanced CVD (PECVD), electron beam evaporation, thermal evaporation, electroplating, sputtering), etching (Reactive Ion Etching (RIE), deep RIE (DRIE), chemical etching in salts and acids and Focused Ion Beam) and lithography (masked lithography, laser lithography and Electron Beam lithography). Experience in additive manufacturing techniques.

Programming Languages: C++, Python, C#

Simulations and Applications: Lumerical, Meep, COMSOL, Ansys, MATLAB, Labview, Visual Studio.

Design: Solidworks, Rhinoceros, AutoCAD, Inventor, KLayout.

CURRENT PROJECTS

New materials for superconductive quantum electrodynamics: *2023-Current* - Growth and test of new materials for superconductive circuits used in the manufacturing of quantum processing units for the development of materials with lower losses and higher critical temperatures.

Principal Investigator: Dr. Boulos Alfakes, TII, UAE.

Design and test of superconductive quantum processing units: *2023-Current* - Quantum electrodynamics design of circuits necessary for the fabrication of multiqubit quantum processing units, including microwave resonators, couplers, coplanar capacitors and tunneling junctions.

Principal Investigator: Dr. Frederio Brito, TII, UAE.

SELECTED COMPLETED PROJECTS

High speed mode and polarization encryption for optical communication networks: *2021-223* - Design of a test bed for a mode / polarization-based modulation and data encryption, classical OPUF based key distribution, and design of devices for photonic chip integration.

Principal Investigator: Dr. Mahmoud Rasras, NYUAD, UAE.

Heterogeneously integrated layered materials for Silicon Photonics: *2021-Current* - Design of photonic systems optimized for heterogeneous integration with layered (2D) materials.

Principal Investigator: Dr. Mahmoud Rasras, NYUAD, UAE.

Quantum Computing Infrastructure: *2023-2024* - Assembly and operation of the infrastructure for test of Quantum Processing Units for Quantum Computing for the first set of quantum computers in the UAE.

Principal Investigators: Prof. Andrew Luthken and Dr. Frederico Brito, TII, UAE.

Plasmonic Enhanced Physically Unclonable Functions: *2020-2021* - Application of plasmonic resonant sites on Silicon photonic structures to generate stochastic responses usable as one-way functions.

Principal Investigator: Prof. Mahmoud Rasras, NYUAD, UAE.

PM2.5 airborne pollutant characterization over the UAE: *2018-2019* - Sample

speciation and source appointment of PM2.5 airborne particulates over the UAE. Definition of recommendations for pollution control to the UAE government agencies and training of their personnel on techniques to measure and record air contaminants.

Direct Manager: Prof. Marouane Temimi, Khalifa University, UAE.

On-board load computers technical specification: 2015-2016 - Development of a classification rule (technical standard) for load and stability computers for inland navigation vessels and electronic systems on board.

Direct Manager: Mr. Jean Michel Chatelier, Bureau Veritas, Belgium.

Naval Datalink System : 2014-2015 - Design, specification and development of prototypes for a new Datalink platform (OSI layers 2-6) for the Colombian Navy.

Direct Manager:: (R) Rear-admiral LH (OF-6) Gustavo Perez, COTECMAR - ARC, Colombia.

Riverine Patrol Craft (LPR-MkII): 2012-2014 Design of updated hull forms, structural layout and new equipment and capacities definition for a rapid Riverine Patrol Craft for the Colombian Navy. Updated hydrodynamic and stability calculations and complete system integration.

Direct Manager: Commander (OF-4) Ricardo Mejía, COTECMAR - ARC, Colombia.

WORK EXPERIENCE

Postdoctoral Researcher

Techonology Innovation Institute

September 2023 - to date

Abu Dhabi

Microwave circuit design and simulation of Quantum Processing Units and operation and maintenance of Class-5 clean-room and dilution refrigerators with combined cooling capacity of 1.65 mW at 20 mK.

Graduate Research Intern

Techonology Innovation Institute

July 2022 - October 2022

Abu Dhabi

Research in the prototyping of Quantum Single photon sources on Silicon Nitride integrated optical chips and Quantum Random Number Generators.

Research Engineer

Khalifa University

August 2018 - April 2019

Abu Dhabi

Research on the measurement, simulation and reporting of environmental data. Sampling and characterization of airborne particulate pollutants using physical (SEM-EDS) and chemical (IC, ICPMS) methods.

Technology and Data Leader

Airborne

January 2018 - June 2020

Abu Dhabi, UAE

Director of the technology development area, developing indoor navigation solutions for people with visual impairments.

Development Engineer

Bureau Veritas

June 2015 - July 2016

Antwerp, BE

Development of rules, regulations, and technical notices for inland navigation vessels and participation in research projects to develop the engineering criteria for the new construction of vessels, notably on onboard electric and electronic systems. Develop training material for courses conducted by the unit.

Marine Designer and Researcher

COTECMAR

February 2012 - June 2015

Cartagena, CO

Engineering design for the Marine Architecture and Electrical and Electronics divi-

sions. Management of engineering teams for the development of small fast marine craft. Research in electronics and telecommunications, developing marine data link protocols and prototypes.

TEACHING EXPERIENCE

I have previous experience teaching in different environments, including training new volunteers at a poverty eradication NGO in Colombia and Belgium, training adult professionals in engineering standards and practices in Belgium, and conducting voluntary Spanish workshops at NYUAD. I have acted as a Teaching Assistant in the following university-level courses:

- *ENGR-UH-1000 Computer Programming for Engineers [Undergraduate]* (Fall-21), Mohammed Eid, New York University, Abu Dhabi
- *MIC630 Fundamentals to Photonics [Graduate]* (Fall-17), Marcus Dahlem, Khalifa University, Abu Dhabi
- *FDN475 Photonic Technology [Undergraduate]* (Spring-18), Marcus Dahlem and Mahmoud Rasras, Khalifa University, Abu Dhabi

CURRENT INTERESTS

System automation and robotics, Integrated Optics, Silicon Photonics, Hardware Security, Cryptography, Additive Manufacturing, Quantum Computing

NON WORK EXPERIENCE

Founding member of EERF (Research Forum for early career scientists at NYUAD), and its Chair in the Spring 2023 semester. Former Member of the research group in Alternative Materials and Processes at the University EIA (Colombia) until 2011, the Nano optics and Optoelectronics research laboratory in Masdar (UAE) until 2020 and the Photonics Research Lab at NYUAD (Abu Dhabi) until 2024. Volunteer (July 2013 - September 2016) with TECHO (NGO based in Chili working on extreme poverty eradication), manager of the Construction and Logistics team in Cartagena, Colombia (December 2014 to July 2015), and National Coordinator of Volunteering in Brussels, Belgium (September 2015 to July 2016). Volleyball, climbing and sailing enthusiast. Rugby player. Member of the Graduate Student Council at NYU Abu Dhabi (from 2019 to 2022). Member of the IEEE Photonics Society, Optica and the Colombian Optical Society.

CONTINUOUS FORMATION

Superconductive Qubit Design, CMC Mixrosystems, Vancouver, Canada (remote) (December 2024). Silicon Photonics Design, CMC Microsystems, Vancouver, Canada (August 2022). Silicon Photonics Design, Fabrication and Data Analysis, UBCx, edx Online Course (November 2019). Teaching Assistant Training, NYUAD, Abu Dhabi, UAE (December 2021). Scientist teaching Sciences, New York Academy of Sciences (August 2020). Train the trainer (Management and development of adult trainings), Opatija, Croatia (March 2016). Inland Navigation Rules and Statutory Regulations, Antwerp, Belgium (November 2015). Solar Energy, DelftX, edx Online Course (December 2015). Electromagnetic Compatibility, Cartagena, Colombia (June 2014).

ACADEMIC REFERENCES

Prof. Dr. Mahmoud Rasras, Associate Professor, New York University, Abu Dhabi, UAE,
mrasras@nyu.edu, +971 (0) 567923399
Prof. Dr. Jaime Viegas, Associate Professor, Khalifa University, Abu Dhabi, UAE.
jaime.viegas@ku.ac.ae, +971 (0) 566986928
Prof. Dr. Frederico Brito, Director Quantum Hardware, Quantum Research Center, TII, Abu Dhabi, UAE.
frederico.brito@tii.ae,

PERSONAL REFERENCES

Dr. Catherine Ramirez, Research Engineer at Syroco, Marseille, France,
lcramirev@gmail.com, +33 665003530

Dr. Dan Cracan, Analog Mixed Signal Verification Engineer, Apple, UK,
dan.cracan@gmail.com, +44 7930 164 762

Dr. Florent Ravaux, Senior Postdoctoral Researcher, TII, Abu Dhabi, UAE,
florent.ravaux@tii.ae, +971 (0) 509009608

PUBLICATIONS IN JOURNALS

Alkhazaleh, A., Villegas, J., Ravaux, F., & Zharinov, A. (2026). *Tantalum-encapsulated niobium superconducting resonators: High internal quality factor and improved temporal stability via surface passivation*. Retrieved from <https://arxiv.org/abs/2604.09050>

Villegas, J. E., Tamalampudi, S. R., & Rasras, M. (2025, Apr). High-speed polarization control in integrated silicon photonics. *Opt. Express*, 33(8), 18477–18491. Retrieved from <https://opg.optica.org/oe/abstract.cfm?URI=oe-33-8-18477> doi: 10.1364/OE.523366

Villegas, J. E., & Rasras, M. (2024, Mar). A compact mode controller (switch) using an inverse-designed 2-mode y-junction. *J. Lightwave Technol.*, 42(6), 2083–2089. Retrieved from <https://opg.optica.org/jlt/abstract.cfm?URI=jlt-42-6-2083>

Tamalampudi, S. R., Villegas, J. E., Dushaq, G., Sankar, R., Paredes, B., & Rasras, M. (2023b). *A high-speed waveguide integrated inse photodetector on sin photonics for nir applications*.

Villegas, J. E., Jimoh, Y. O., & Rasras, M. (2023). Polarization encryption system using commercial lcds for additive manufacturing. *Opt. Continuum*, 2(4), 783–792. doi: 10.1364/OPTCON.480991

Villegas, J. E., Paredes, B., & Rasras, M. (2021a). Experimental studies of plasmonics-enhanced optical physically unclonable functions. *Opt. Express*, 29(20), 32020–32030. doi: 10.1364/OE.437636

Tamalampudi, S. R., Villegas, J. E., Dushaq, G., Sankar, R., Paredes, B., & Rasras, M. (2023a). High-speed waveguide-integrated inse photodetector on sin photonics for near-infrared applications. *Advanced Photonics Research*, 4(11), 2300162. doi: 10.1002/adpr.202300162

Zafar, H., Paredes, B., Villegas, J., Rasras, M., & Pereira, M. F. (2023, Jun). O-band te- and tm-mode densely packed adiabatically bent waveguide arrays on the silicon-on-insulator platform. *Opt. Express*, 31(13), 21389–21398. doi: 10.1364/OE.493077

Zafar, H., Paredes, B., Taha, I., Villegas, J. E., Rasras, M., & Pereira, M. F. (2023). Compact and broadband adiabatically bent superlattice-waveguides with negligible insertion loss and ultra-low crosstalk. *IEEE Journal of Selected Topics in Quantum Electronics*, 29(6: Photonic Signal Processing), 1-9. doi: 10.1109/JSTQE.2023.3241617

- Tamalampudi, S. R., Dushaq, G., Villegas, J. E., Paredes, B., & Rasras, M. S. (2023). A multi-layered gage electro-optic device integrated in silicon photonics. *Journal of Lightwave Technology*, 1-7. doi: 10.1109/JLT.2023.3237818
- Dushaq, G., Villegas, J. E., Paredes, B., Tamalampudi, S. R., & Rasras, M. S. (2023a). Anisotropic van der waals 2d geas integrated on silicon four-waveguide crossing. *J. of Lightwave Technology*, 41(6), 1784-1789. doi: 10.1109/JLT.2022.3229069
- Tamalampudi, S. R., Dushaq, G., Villegas, J. E., Paredes, B., & Rasras, M. S. (2022). Si2te3 photodetectors for optoelectronic integration at telecommunication wavelengths. *IEEE Journal of Selected Topics in Quantum Electronics*, 28(3), 1-7. doi: 10.1109/JSTQE.2021.3126612
- Zafar, H., Zhai, Y., Villegas, J. E., Ravaux, F., Kennedy, K. L., Pereira, M. F., ... Anjum, D. H. (2022, Mar). Compact broadband (o, e, s, c, l & u bands) silicon te-pass polarizer based on ridge waveguide adiabatic s-bends. *Opt. Express*, 30(6), 10087–10095. doi: 10.1364/OE.452823
- Dushaq, G., Paredes, B., Villegas, J. E., Tamalampudi, S. R., & Rasras, M. (2022b, May). On-chip integration of 2d van der waals germanium phosphide (gep) for active silicon photonics devices. *Opt. Express*, 30(10), 15986–15997. doi: 10.1364/OE.457242
- Paredes, B., Mohammed, Z., Villegas, J., & Rasras, M. (2021a). Dual-band (o and c-bands) two-mode multiplexer on the soi platform. *IEEE Photonics Journal*, 13(3), 1-9. doi: 10.1109/JPHOT.2021.3075292
- Tamalampudi, S. R., Dushaq, G., Villegas, J. E., Rajput, N. S., Paredes, B., Elamurugu, E., & Rasras, M. S. (2021). Short-wavelength infrared (swir) photodetector based on multi-layer 2d gage. *Opt. Express*, 29(24), 39395–39405. doi: 10.1364/OE.442845
- Alfakes, B., Garlisi, C., Villegas, J., Al-Hagri, A., Tamalampudi, S., Rajput, N. S., ... Chiesa, M. (2020). Enhanced photoelectrochemical performance of atomic layer deposited hf-doped zno. *Surface and Coatings Technology*, 385, 125352. doi: 10.1016/j.surfcoat.2020.125352
- Odeh, M., Twayana, K., Sloyan, K., Villegas, J. E., Chandran, S., & Dahlem, M. S. (2019). Mode sensitivity analysis of subwavelength grating slot waveguides. *IEEE Photonics Journal*, 11(5), 1-10. doi: 10.1109/JPHOT.2019.2939088
- Alfakes, B., Villegas, J., Apostoleris, H., Devarapalli, R., Tamalampudi, S., Lu, J.-Y., ... Chiesa, M. (2019). Optoelectronic tunability of hf-doped zno for photovoltaic applications. *The Journal of Physical Chemistry C*, 123(24), 15258-15266. doi: 10.1021/acs.jpcc.9b02253

CONFERENCE PROCEEDINGS

- Villegas, J. E., Tamalampudi, S. R., & Rasras, M. S. (2024). A high-speed compact polarization controller in silicon photonics. In *2024 optical fiber communications conference and exhibition (ofc)* (p. 1-3).

- Dushaq, G., Villegas, J. E., Paredes, B., Tamalampudi, S. R., & Rasras, M. (2023b). Four-waveguide crossing functions utilizing anisotropic van der waals 2d geas. In *Cleo 2023* (p. 1-2). Optica.
- Tamalampudi, S. R., Villegas, J. E., Paredes, B., Dushaq, G., & Rasras, M. S. (2023). Integrated sin/inse waveguide photodetector for o-band applications. In *Cleo: Science and innovations* (p. JW2A. 41). Optica.
- Villegas, J. E., & Rasras, M. (2023). Multimode silicon y-splitter with in-phase and off-phase te0 outputs. In *2023 ieee silicon photonics conference (siphotonics)* (p. 1-2). doi: 10.1109/SiPhotonics55903.2023.10141955
- Tamalampudi, S., Dushaq, G., Villegas, J., & Rasras, M. (2023). Integrated 2d gagete electro-optic phase shifter in silicon photonics. In *International conference on meta-materials, photonic crystals and plasmonics* (pp. 227–229).
- Paredes, B., Mohammed, Z., Villegas, J. E., & Rasras, M. (2021c). Ultra-compact ultra-broadband two-mode transverse-electric based swg multiplexer demonstrated at 64 gbps. In *Ecoc* (p. 1-3). Optica.
- Villegas, J. E., Dushaq, G., & Rasras, M. (2022). Inverse design of a 4-way crossing for integration of anisotropic materials on silicon waveguides. In *Frontiers in optics + laser science 2022 (fio, ls)* (p. JTu5A.58). Optica Publishing Group.
- Dushaq, G., Paredes, B., Villegas, J. E., Tamalampudi, S. R., & Rasras, M. (2022a). Hybrid integration of multilayers gep on a silicon photonics platform for optoelectronic application. In *Conference on lasers and electro-optics* (p. JW3A.33). Optica Publishing Group.
- Paredes, B., Mohammed, Z., Villegas, J., & Rasras, M. (2021b). An ultra-broadband two-mode transverse-electric multiplexer in soi platform. In *2021 optical fiber communications conference and exhibition (ofc)* (p. 1-3).
- Villegas, J. E., Paredes, B., & Rasras, M. (2021b). Performance of silicon opufs under variable input losses. In *2021 conference on lasers and electro-optics europe & european quantum electronics conference (cleo/europe-eqec)* (p. 1-1). doi: 10.1109/CLEO/Europe-EQEC52157.2021.9542040
- Mohammed, Z., Paredes, B., Villegas, J., & Rasras, M. (2021). An ultra-compact cmos compatible mmi based 1310/1550 nm wavelength (de) multiplexer. In *2021 european conference on optical communication (ecoc)* (p. 1-3). doi: 10.1109/ECOC52684.2021.9606089
- Odeh, M., Twayana, K., Sloyan, K., Villegas, J. E., Chandran, S., & Dahlem, M. S. (2018). Slotted subwavelength grating waveguides for compact optofluidic sensors. In *2018 international conference on optical mems and nanophotonics (omn)* (p. 1-2). doi: 10.1109/OMN.2018.8454591
- Villegas, J. E., Odeh, M., & Dahlem, M. S. (2017). Integrated gradient-index planar optofluidic polymer waveguide. In *Frontiers in optics 2017* (p. JTu2A.18). Optica Publishing Group.